

Figure 2.46 Capacitance-voltage characteristic of an NMOS device.

References

[1] R. S. Muller and T. I. Kamins, *Device Electronics for Integrated Circuits*, 2nd ed. (New York: Wiley, 1986).
[2] Y. Tsividis, *Operation and Modeling of the MOS Transistor*, 2nd ed. (Boston: McGraw-Hill, 1999).
[3] Y. Taur and T. H. Ning, *Fundamentals of Modern VLSI Devices* (New York: Cambridge University Press, 1998).

Problems

Unless otherwise stated, in the following problems, use the device data shown in Table 2.1 and assume $V_{DD} = 3$ V where necessary.

2.1. For $W/L = 50/0.5$, plot the drain current of an NFET and a PFET as a function of $|V_{GS}|$ as $|V_{GS}|$ varies from 0 to 3 V. Assume that $|V_{DS}| = 3$ V.

2.2. For $W/L = 50/0.5$ and $|I_D| = 0.5$ mA, calculate the transconductance and output impedance of both NMOS and PMOS devices. Also, find the "intrinsic gain," defined as $g_m r_O$.

2.3. Derive expressions for $g_m r_O$ in terms of I_D and W/L. Plot $g_m r_O$ as a function of I_D with L as a parameter. Note that $\lambda \propto 1/L$.

2.4. Plot I_D versus V_{GS} for a MOS transistor (a) with V_{DS} as a parameter, and (b) with V_{BS} as a parameter. Identify the break points in the characteristics.

2.5. Sketch I_X and the transconductance of the transistor as a function of V_X for each circuit in Fig. 2.47 as V_X varies from 0 to V_{DD}. In part (a), assume that V_X varies from 0 to 1.5 V.

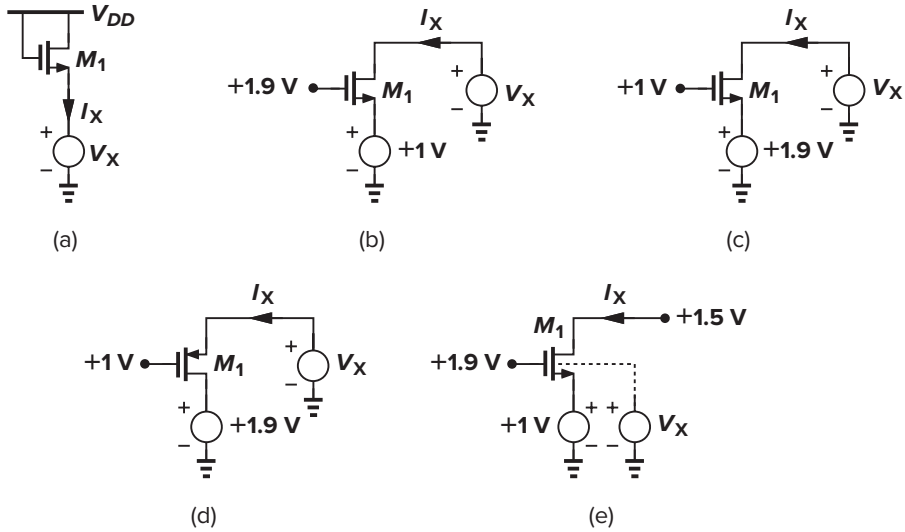


Figure 2.47

2.6. Sketch I_X and the transconductance of the transistor as a function of V_X for each circuit in Fig. 2.48 as V_X varies from 0 to V_{DD}.

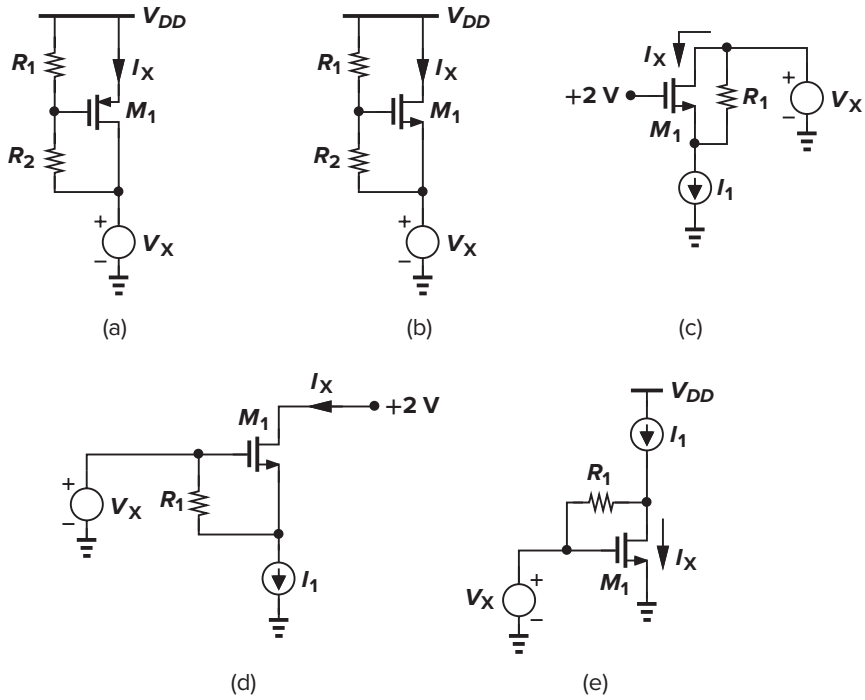


Figure 2.48

2.7. Sketch V_{out} as a function of V_{in} for each circuit in Fig. 2.49 as V_{in} varies from 0 to V_{DD}.

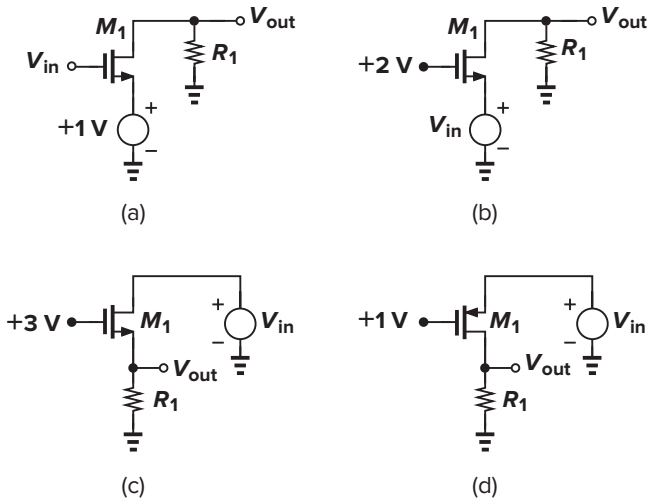


Figure 2.49

2.8. Sketch V_{out} as a function of V_{in} for each circuit in Fig. 2.50 as V_{in} varies from 0 to V_{DD}.

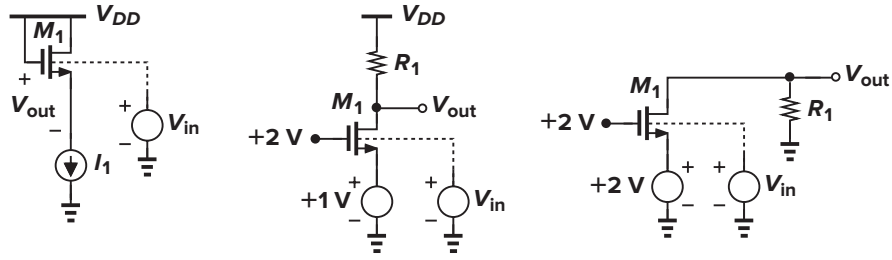


Figure 2.50

2.9. Sketch V_X and I_X as a function of time for each circuit in Fig. 2.51. The initial voltage of C_1 is equal to 3 V. In part (e), assume that the switch turns off at $t = 0$.

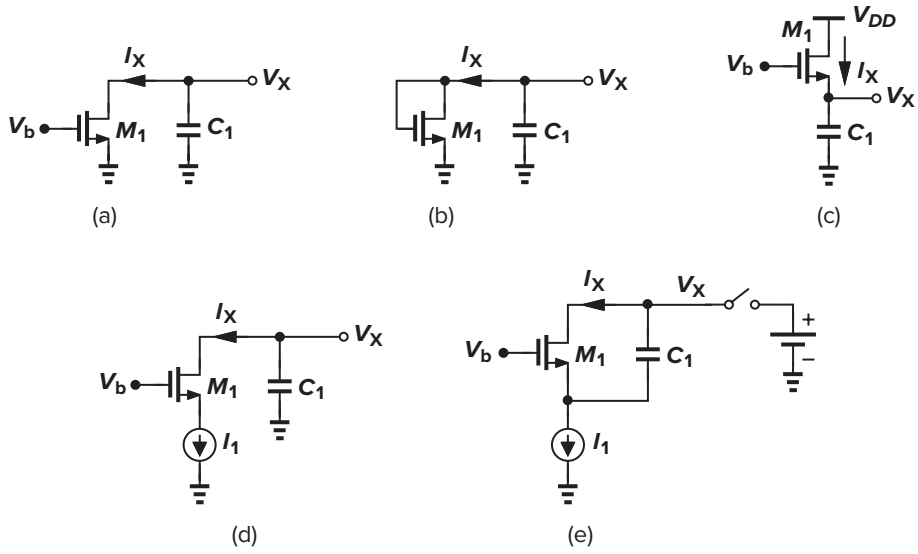


Figure 2.51

2.10. Sketch V_X and I_X as a function of time for each circuit in Fig. 2.52. The initial voltage of each capacitor is shown.

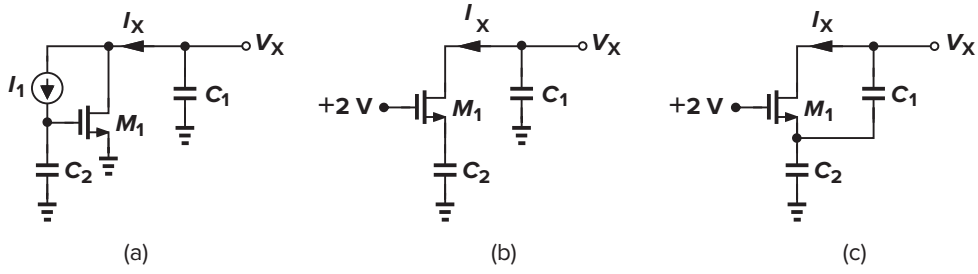


Figure 2.52

2.11. Sketch V_X as a function of time for each circuit in Fig. 2.53. The initial voltage of each capacitor is shown.

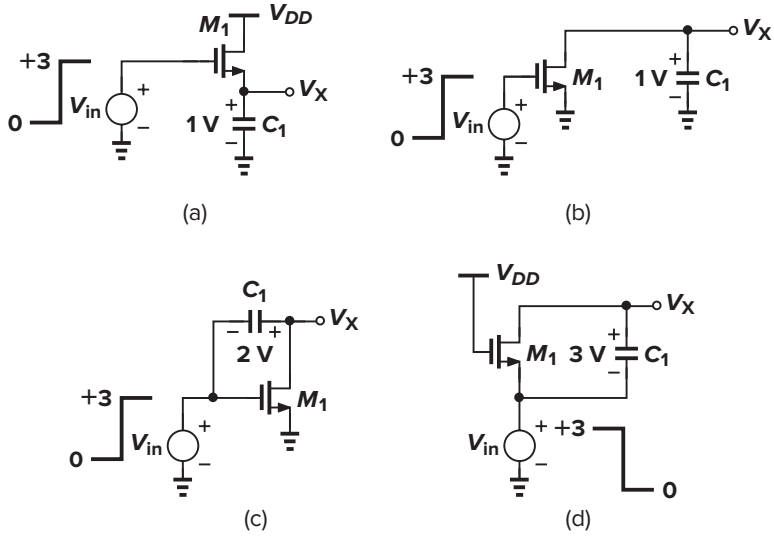


Figure 2.53

2.12. Sketch V_X as a function of time for each circuit in Fig. 2.54. The initial voltage of each capacitor is shown.

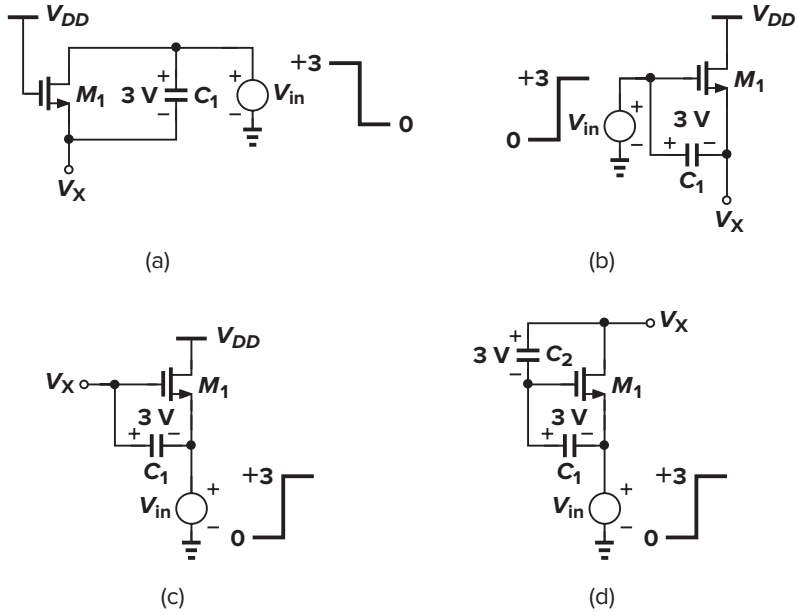


Figure 2.54

2.13. The transit frequency, f_T, of a MOSFET is defined as the frequency at which the small-signal current gain of the device drops to unity while the source and drain terminals are held at ac ground.
(a) Prove that

$$f_T = \frac{g_m}{2\pi(C_{GD} + C_{GS})} \tag{2.55}$$

Note that f_T does not include the effect of the S/D junction capacitance.
(b) Suppose the gate resistance, R_G, is significant and the device is modeled as a distributed set of n transistors, each with a gate resistance equal to R_G/n. Prove that the f_T of the device is independent of R_G and still equal to the value given above.
(c) For a given bias current, the minimum allowable drain-source voltage for operation in saturation can be reduced only by increasing the width and hence the capacitances of the transistor. Using square-law characteristics, prove that

$$f_T = \frac{\mu_n}{2\pi} \frac{V_{GS} - V_{TH}}{L^2} \tag{2.56}$$

This relation indicates how the speed is limited as a device is designed to operate with lower supply voltages.

2.14. Calculate the f_T of a MOS device in the subthreshold region and compare the result with that obtained in Prob. 2.13.

2.15. For a saturated NMOS device having $W = 50\ \mu$m and $L = 0.5\ \mu$m, calculate all the capacitances. Assume that the minimum (lateral) dimension of the S/D areas is $1.5\ \mu$m and that the device is folded as shown in Fig. 2.33(b). What is the f_T if the drain current is 1 mA?

2.16. Consider the structure shown in Fig. 2.55. Determine I_D, as a function of V_{GS} and V_{DS}, and prove that the structure can be viewed as a single transistor having an aspect ratio $W/(2L)$. Assume that $\lambda = \gamma = 0$.

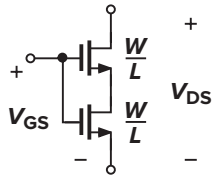


Figure 2.55

2.17. For an NMOS device operating in saturation, plot W/L versus $V_{GS} - V_{TH}$ if (a) I_D is constant, and (b) g_m is constant.

2.18. Explain why the structures shown in Fig. 2.56 cannot operate as current sources even though the transistors are in saturation.

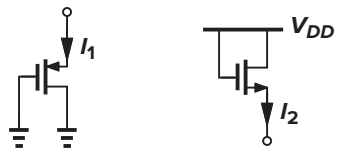


Figure 2.56

2.19. Considering the body effect as "back-gate effect," explain intuitively why γ is directly proportional to $\sqrt{N_{sub}}$ and inversely proportional to C_{ox}.

2.20. A "ring" MOS structure is shown in Fig. 2.57. Explain how the device operates and estimate its equivalent aspect ratio. Compare the drain junction capacitance of this structure with that of the devices shown in Fig. 2.33.

2.21. Suppose we have received an NMOS transistor in a package with four unmarked pins. Describe the minimum number of dc measurement steps using an ohmmeter that is necessary to determine the gate, source/drain, and bulk terminals of the device.

2.22. Repeat Prob. 2.21 if the type of the device (NFET or PFET) is not known.

2.23. For an NMOS transistor, the threshold voltage is known, but $\mu_n C_{ox}$ and W/L are not. Assume that $\lambda = \gamma = 0$. If we cannot measure C_{ox} independently, is it possible to devise a sequence of dc measurement tests to determine $\mu_n C_{ox}$ and W/L? What if we have two transistors and we know that one has twice the aspect ratio of the other?

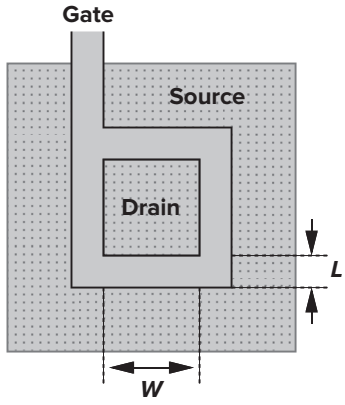


Figure 2.57

2.24. Sketch I_X versus V_X for each of the composite structures shown in Fig. 2.58 with V_G as a parameter. Also, sketch the equivalent transconductance. Assume that $\lambda = \gamma = 0$.

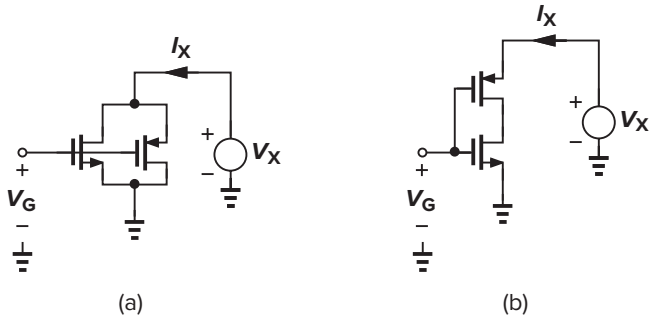


Figure 2.58

2.25. An NMOS current source with $I_D = 0.5$ mA must operate with drain-source voltages as low as 0.4 V. If the minimum required output impedance is 20 kΩ, determine the width and length of the device. Calculate the gate-source, gate-drain, and drain-substrate capacitance if the device is folded as in Fig. 2.33 and $E = 3$ μm.

2.26. Consider the circuit shown in Fig. 2.59, where the initial voltage at node X is equal to V_{DD}. Assuming that $\lambda = \gamma = 0$ and neglecting other capacitances, plot V_X and V_Y versus time if (a) V_{in} is a positive step with amplitude $V_0 > V_{TH}$, and (b) V_{in} is a negative step with amplitude $V_0 = V_{TH}$.

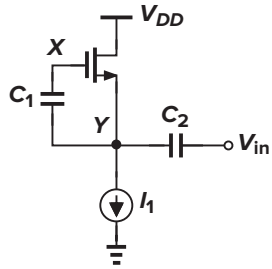


Figure 2.59

2.27. An NMOS device operating in the subthreshold region has a ζ of 1.5. What variation in V_{GS} results in a tenfold change in I_D? If $I_D = 10$ μA, what is g_m?

2.28. Consider an NMOS device with $V_G = 1.5$ V and $V_S = 0$. Explain what happens if we continually decrease V_D below zero or increase V_{sub} above zero.

2.29. Consider the arrangement shown in Fig. 2.60. Explain what happens to the pinch-off point as V_G increases.

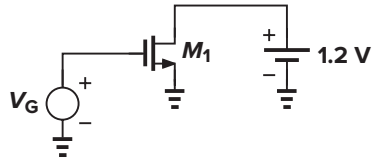


Figure 2.60

2.30. From Fig. 2.20, plot I_D vs. $V_{GS} - V_{TH}$ if W/L is constant, $V_{GS} - V_{TH}$ vs. I_D if W/L is constant, and W/L vs. $V_{GS} - V_{TH}$ if I_D is constant.

2.31. Plotted in Fig. 2.61 are the charactersitics of a square-law NMOS device with $W/L_{drawn} = 5\ \mu$m/40 nm and $t_{ox} = 18$ Å. Here, V_{GS} is incremented in equal steps. Estimate μ_n, V_{TH}, λ, and the V_{GS} steps.

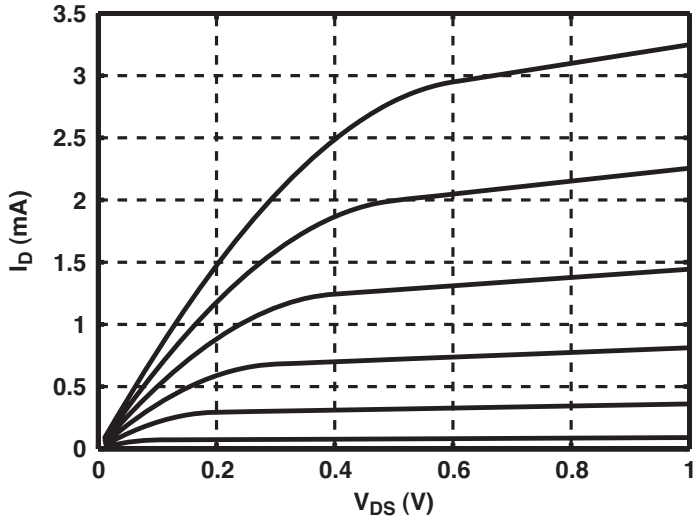


Figure 2.61

FETs operating in saturation) for all transistors. If the drain of a device is connected to a high impedance (e.g., the drain of another), then add r_O to its model. At this point, the basic properties of most circuits can be determined by inspection. In a second, more accurate iteration, the body effect of devices whose source or bulk is not at ac ground can be included as well.

For bias calculations, it is usually adequate to neglect channel-length modulation and body effect in the first pass. These effects do introduce some error, but they can be included in the next iteration step—after the basic properties are understood.

In today's analog design, simulation of circuits is essential because the behavior of short-channel MOSFETs cannot be predicted accurately by hand calculations. Nonetheless, if the designer avoids a simple and intuitive analysis of the circuit and hence skips the task of gaining insight, then he/she cannot interpret the simulation results intelligently. For this reason, we say, "Don't let the computer think for you." Some say, "Don't be a SPICE monkey."

Problems

Unless otherwise stated, in the following problems, use the device data shown in Table 2.1 and assume that $V_{DD} = 3$ V where necessary. All device dimensions are effective values and in microns.

3.1. For the circuit of Fig. 3.13, calculate the small-signal voltage gain if $(W/L)_1 = 50/0.5$, $(W/L)_2 = 10/0.5$, and $I_{D1} = I_{D2} = 0.5$ mA. What is the gain if M_2 is implemented as a diode-connected PMOS device (Fig. 3.16)?

3.2. In the circuit of Fig. 3.18, assume that $(W/L)_1 = 50/0.5$, $(W/L)_2 = 50/2$, and $I_{D1} = I_{D2} = 0.5$ mA when both devices are in saturation. Recall that $\lambda \propto 1/L$.
(a) Calculate the small-signal voltage gain.
(b) Calculate the maximum output voltage swing while both devices are saturated.

3.3. In the circuit of Fig. 3.4(a), assume that $(W/L)_1 = 50/0.5$, $R_D = 2\,\text{k}\Omega$, and $\lambda = 0$.
(a) What is the small-signal gain if M_1 is in saturation and $I_D = 1$ mA?
(b) What input voltage places M_1 at the edge of the triode region? What is the small-signal gain under this condition?
(c) What input voltage drives M_1 into the triode region by 50 mV? What is the small-signal gain under this condition?

3.4. Suppose the common-source stage of Fig. 3.4(a) is to provide an output swing from 1 V to 2.5 V. Assume that $(W/L)_1 = 50/0.5$, $R_D = 2\,\text{k}\Omega$, and $\lambda = 0$.
(a) Calculate the input voltages that yield $V_{out} = 1$ V and $V_{out} = 2.5$ V.
(b) Calculate the drain current and the transconductance of M_1 for both cases.
(c) How much does the small-signal gain, $g_m R_D$, vary as the output goes from 1 V to 2.5 V? (Variation of small-signal gain can be viewed as nonlinearity.)

3.5. Calculate the intrinsic gain of an NMOS device and a PMOS device operating in saturation with $W/L = 50/0.5$ and $|I_D| = 0.5$ mA. Repeat these calculations if $W/L = 100/1$.

3.6. Assuming a constant L, plot the intrinsic gain of a satuated device versus the gate-source voltage if **(a)** the drain current is constant, **(b)** W is constant.

3.7. Assuming a constant L, plot the intrinsic gain of a saturated device versus W/L if **(a)** the gate-source voltage is constant, **(b)** the drain current is constant.

3.8. An NMOS transistor with $W/L = 50/0.5$ is biased with $V_G = +1.2$ V and $V_S = 0$. The drain voltage is varied from 0 to 3 V.
(a) Assuming the bulk voltage is zero, plot the intrinsic gain versus V_{DS}.
(b) Repeat part **(a)** for a bulk voltage of -1 V.

3.9. For an NMOS device operating in saturation, plot g_m, r_O, and $g_m r_O$ as the bulk voltage goes from 0 to $-\infty$ while other terminal voltages remain constant.

3.10. Consider the circuit of Fig. 3.13 with $(W/L)_1 = 50/0.5$ and $(W/L)_2 = 10/0.5$. Assume that $\lambda = \gamma = 0$.
(a) At what input voltage is M_1 at the edge of the triode region? What is the small-signal gain under this condition?
(b) What input voltage drives M_1 into the triode region by 50 mV? What is the small-signal gain under this condition?

3.11. Repeat Problem 3.10 if body effect is not neglected.

3.12. In the circuit of Fig. 3.17, $(W/L)_1 = 20/0.5$, $I_1 = 1$ mA, and $I_S = 0.75$ mA. Assuming $\lambda = 0$, calculate $(W/L)_2$ such that M_1 is at the edge of the triode region. What is the small-signal voltage gain under this condition?

3.13. Plot the small-signal gain of the circuit shown in Fig. 3.17 as I_S goes from 0 to $0.75I_1$. Assume that M_1 is always saturated, and neglect channel-length modulation and body effect.

3.14. The circuit of Fig. 3.18 is designed to provide an output voltage swing of 2.2 V with a bias current of 1 mA and a small-signal voltage gain of 100. Calculate the dimensions of M_1 and M_2.

3.15. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.78 as V_{in} varies from 0 to V_{DD}. Identify important transition points.

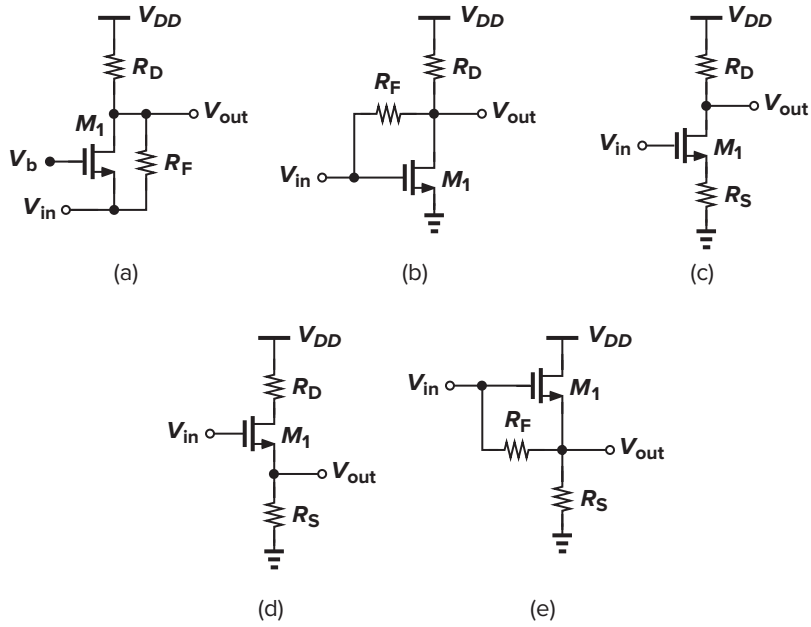


Figure 3.78

3.16. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.79 as V_{in} varies from 0 to V_{DD}. Identify important transition points.

3.17. Sketch V_{out} versus V_{in} for the circuits of Fig. 3.80 as V_{in} varies from 0 to V_{DD}. Identify important transition points.

3.18. Sketch I_X versus V_X for the circuits of Fig. 3.81 as V_X varies from 0 to V_{DD}. Identify important transition points.

3.19. Sketch I_X versus V_X for the circuits of Fig. 3.82 as V_X varies from 0 to V_{DD}. Identify important transition points.

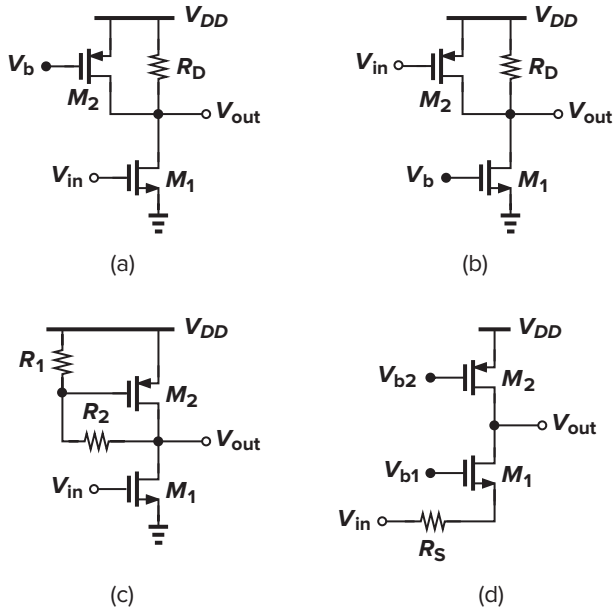


Figure 3.79

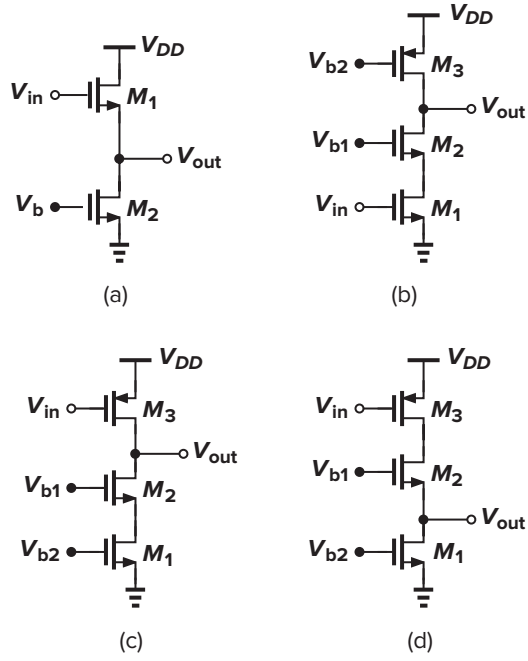
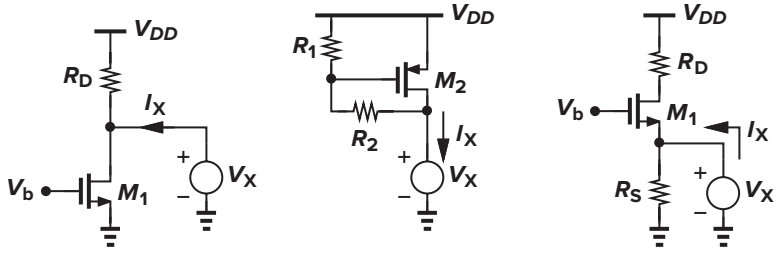
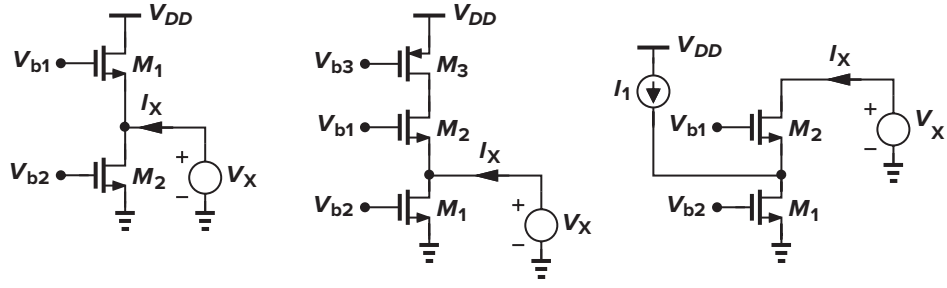


Figure 3.80

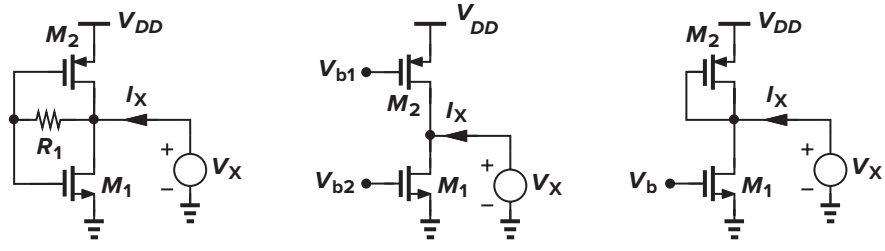


(a) (b) (c)

Figure 3.81



(a) (b) (c)



(d) (e) (f)

Figure 3.82

3.20. Assuming all MOSFETs are in saturation, calculate the small-signal voltage gain of each circuit in Fig. 3.83 ($\lambda \neq 0, \gamma = 0$).

3.21. Assuming all MOSFETs are in saturation, calculate the small-signal voltage gain of each circuit in Fig. 3.84 ($\lambda \neq 0, \gamma = 0$).

3.22. Sketch V_X and V_Y as a function of time for each circuit in Fig. 3.85. The initial voltage across C_1 is equal to V_{DD}.

3.23. In the cascode stage of Fig. 3.59, assume that $(W/L)_1 = 50/0.5$, $(W/L)_2 = 10/0.5$, $I_{D1} = I_{D2} = 0.5$ mA, and $R_D = 1$ kΩ.

(a) Choose V_b such that M_1 is 50 mV away from the triode region.

(b) Calculate the small-signal voltage gain.

(c) Using the value of V_b found in part (a), calculate the maximum output voltage swing. Which device enters the triode region first as V_{out} falls?

(d) Calculate the swing at node X for the maximum output swing obtained above.

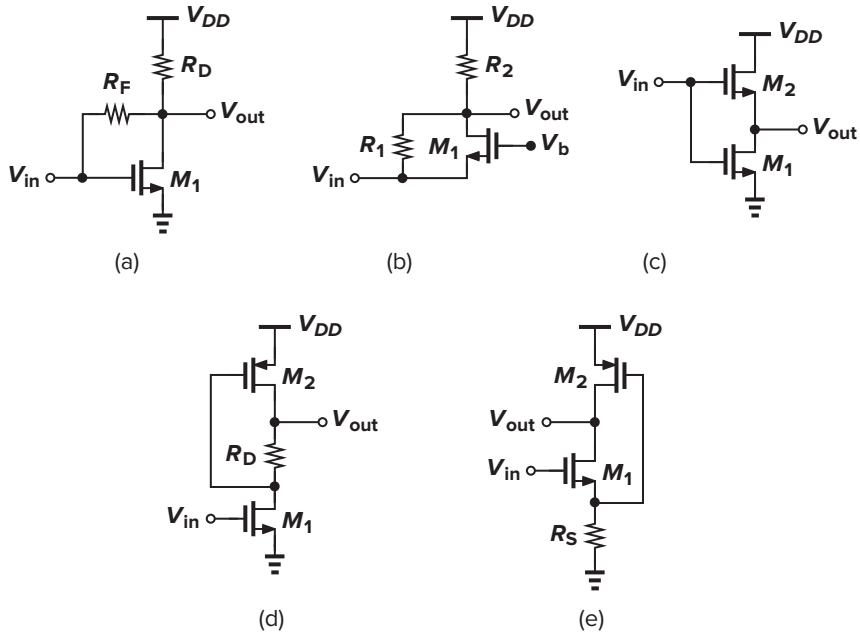


Figure 3.83

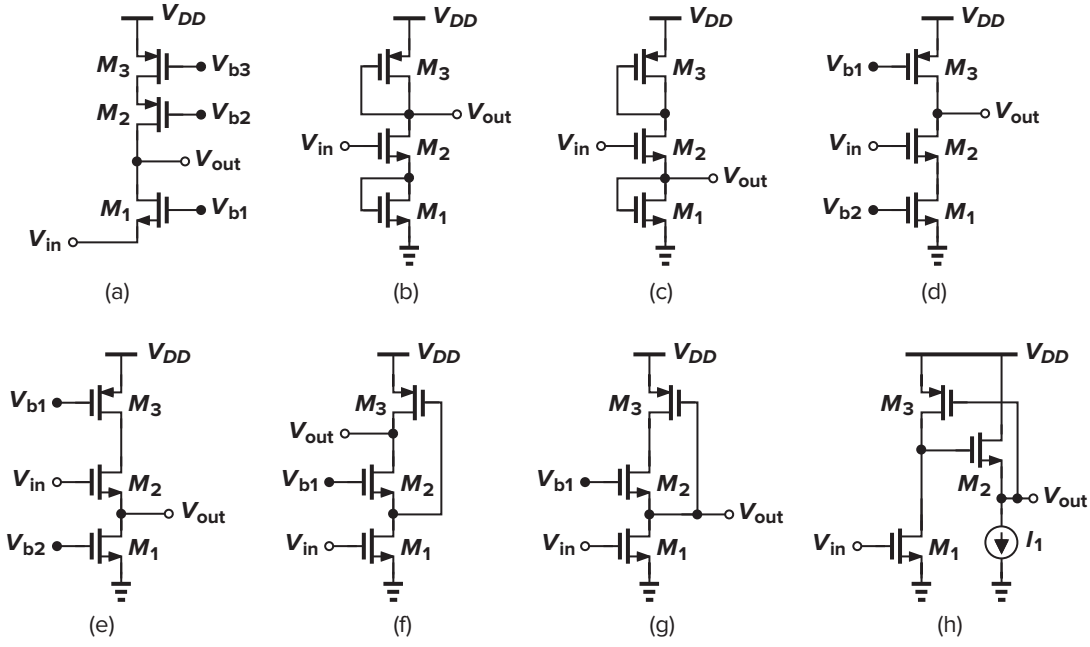


Figure 3.84

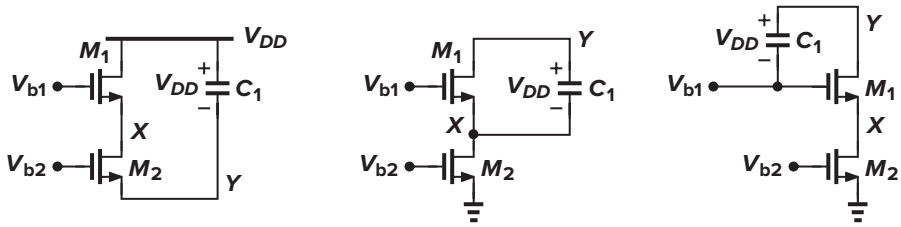


Figure 3.85

3.24. Consider the circuit of Fig. 3.23 with $(W/L)_1 = 50/0.5$, $R_D = 2\ \text{k}\Omega$, and $R_S = 200\ \Omega$.
(a) Calculate the small-signal voltage gain if $I_D = 0.5$ mA.
(b) Assuming $\lambda = \gamma = 0$, calculate the input voltage that places M_1 at the edge of the triode region. What is the gain under this condition?

3.25. Suppose the circuit of Fig. 3.22 is designed for a voltage gain of 5. If $(W/L)_1 = 20/0.5$, $I_{D1} = 0.5$ mA, and $V_b = 0$ V:
(a) Calculate the aspect ratio of M_2.
(b) What input level places M_1 at the edge of the triode region? What is the small-signal gain under this condition?
(c) What input level places M_2 at the edge of the saturation region? What is the small-signal gain under this condition?

3.26. Sketch the small-signal voltage gain of the circuit shown in Fig. 3.22 as V_b varies from 0 to V_{DD}. Consider two cases:
(a) M_1 enters the triode region before M_2 is saturated;
(b) M_1 enters the triode region after M_2 is saturated.

3.27. A source follower can operate as a level shifter. Suppose the circuit of Fig. 3.37(b) is designed to shift the voltage level by 1 V, i.e., $V_{in} - V_{out} = 1$ V.
(a) Calculate the dimensions of M_1 and M_2 if $I_{D1} = I_{D2} = 0.5$ mA, $V_{GS2} - V_{GS1} = 0.5$ V, and $\lambda = \gamma = 0$.
(b) Repeat part (a) if $\gamma = 0.45\ \text{V}^{-1}$ and $V_{in} = 2.5$ V. What is the minimum input voltage for which M_2 remains saturated?

3.28. Sketch the small-signal gain, V_{out}/V_{in}, of the cascode stage shown in Fig. 3.59 as V_b goes from 0 to V_{DD}. Assume that $\lambda = \gamma = 0$.

3.29. The cascode of Fig. 3.70 is designed to provide an output swing of 1.9 V with a bias current of 0.5 mA. If $\gamma = 0$ and $(W/L)_{1-4} = W/L$, calculate V_{b1}, V_{b2}, and W/L. What is the voltage gain if $L = 0.5\ \mu$m?

3.30. Consider the gate-source voltage of M_1 in Fig. 3.23(a): $V_{GS} = V_{in} - I_D R_S$. Determine ΔV_{GS} in response to a change in V_{in} and show that it *decreases* as $g_m R_S$ increases. How does this trend show that the circuit becomes more linear?

3.31. Prove that the voltage gain from V_{DD} to V_{out} in Fig. 3.21 is given by

$$\frac{V_{out}}{V_{in}} = \frac{g_{m2} r_{O2} + 1}{r_{O2} + r_{O1}} r_{O1} \tag{3.147}$$

3.32. In the circuit shown in Fig. 3.86, prove that

$$\frac{V_{out1}}{V_{out2}} = \frac{-R_D}{R_S} \tag{3.148}$$

where V_{out1} and V_{out2} are small-signal quantities and $\lambda,\ \gamma > 0$.

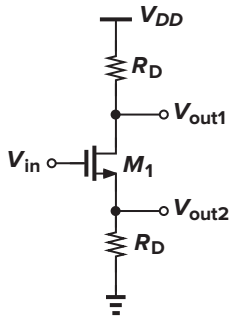


Figure 3.86

3.33. The CG stage of Fig. 3.51(a) is designed such that its input resistance (seen at node X) matches the signal source resistance, R_S. If $\lambda,\ \gamma > 0$, prove that

$$\frac{V_{out}}{V_{in}} = \frac{1 + (g_m + g_{mb})r_O}{2 + \left(1 + \dfrac{r_O}{R_D}\right)} \tag{3.149}$$

Also, prove that

$$\frac{V_{out}}{V_{in}} = \frac{R_D}{2R_S} \tag{3.150}$$

3.34. Calculate the voltage gain of a source follower using the lemma $A_v = -G_m R_{out}$. Assume that the circuit drives a load resistance of R_L and $\lambda,\ \gamma > 0$.

3.35. Calculate the voltage gain of a common-gate stage using the lemma $A_v = -G_m R_{out}$. Assume a source resistance of R_S and $\lambda,\ \gamma > 0$.

3.36. How many amplifier topologies can you create using each of the structures shown in Fig. 3.87 and no other transistors? (The source and drain terminals can be swapped.)

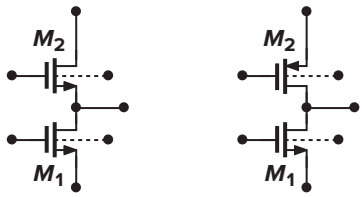


Figure 3.87

References

[1] P. R. Gray and R. G. Meyer, *Analysis and Design of Analog Integrated Circuits*, 3d ed. (New York: Wiley, 1993).
[2] B. Gilbert, "A Precise Four-Quadrant Multiplier with Subnanosecond Response," *IEEE J. Solid-State Circuits*, vol. SC-3, pp. 365–373, Dec. 1968.

Problems

Unless otherwise stated, in the following problems, use the device data shown in Table 2.1 and assume that $V_{DD} = 3$ V where necessary. All device dimensions are effective values and in microns.

4.1. Suppose the total capacitance between adjacent lines in Fig. 4.2 is 10 fF and the capacitance from the drains of M_1 and M_2 to ground is 100 fF.
(a) What is the amplitude of the glitches in the analog output in Fig. 4.2(a) for a clock swing of 3 V?
(b) If in Fig. 4.2(b), the capacitance between L_1 and L_2 is 10% less than that between L_1 and L_3, what is the amplitude of the glitches in the differential analog output for a clock swing of 3 V?

4.2. Sketch the small-signal differential voltage gain of the circuit shown in Fig. 4.9(a) if V_{DD} varies from 0 to 3 V. Assume that $(W/L)_{1-3} = 50/0.5$, $V_{in,CM} = 1.3$ V, and $V_b = 1$ V.

4.3. Construct the plots of Fig. 4.9(c) for a differential pair using PMOS transistors.

4.4. In the circuit of Fig. 4.11, $(W/L)_{1,2} = 50/0.5$ and $I_{SS} = 0.5$ mA.
(a) What is the maximum allowable output voltage swing if $V_{in,CM} = 1.2$ V?
(b) What is the voltage gain under this condition?

4.5. A differential pair uses input NMOS devices with $W/L = 50/0.5$ and a tail current of 1 mA.
(a) What is the equilibrium overdrive voltage of each transistor?
(b) How is the tail current shared between the two sides if $V_{in1} - V_{in2} = 50$ mA?
(c) What is the equivalent G_m under this condition?
(d) For what value of $V_{in1} - V_{in2}$ does the G_m drop by 10%? By 90%?

4.6. Repeat Problem 4.5 with $W/L = 25/0.5$ and compare the results.

4.7. Repeat Problem 4.5 with a tail current of 2 mA and compare the results.

4.8. Sketch I_{D1} and I_{D2} in Fig. 4.19 versus $V_{in1} - V_{in2}$. For what value of $V_{in1} - V_{in2}$ are the two currents equal?

4.9. Consider the circuit of Fig. 4.32, assuming $(W/L)_{1,2} = 50/0.5$ and $R_D = 2$ kΩ. Suppose R_{SS} represents the output impedance of an NMOS current source with $(W/L)_{SS} = 50/0.5$ and a drain current of 1 mA. The input signal consists of $V_{in,DM} = 10$ mV$_{pp}$ and $V_{in,CM} = 1.5$ V $+ V_n(t)$, where $V_n(t)$ denotes noise with a peak-to-peak amplitude of 100 mV. Assume that $\Delta R/R = 0.5\%$.
(a) Calculate the output differential signal-to-noise ratio, defined as the signal amplitude divided by the noise amplitude.
(b) Calculate the CMRR.

4.10. Repeat Problem 4.9 if $\Delta R = 0$, but M_1 and M_2 suffer from a threshold voltage mismatch of 1 mV.

4.11. Suppose the differential pair of Fig. 4.37(a) is designed with $(W/L)_{1,2} = 50/0.5$, $(W/L)_{3,4} = 10/0.5$, and $I_{SS} = 0.5$ mA. Also, I_{SS} is implemented with an NMOS device having $(W/L)_{SS} = 50/0.5$.
(a) What are the minimum and maximum allowable input CM levels if the differential swings at the input and output are small?
(b) For $V_{in,CM} = 1.2$ V, sketch the small-signal differential voltage gain as V_{DD} goes from 0 to 3 V.

4.12. In Problem 4.11, suppose M_1 and M_2 have a threshold voltage mismatch of 1 mV. What is the CMRR?

4.13. In Problem 4.11, suppose $W_3 = 10$ μm, but $W_4 = 11$ μm. Calculate the CMRR.

4.14. For the differential pairs of Fig. 4.37(a) and (b), calculate the differential voltage gain if $I_{SS} = 1$ mA, $(W/L)_{1,2} = 50/0.5$, and $(W/L)_{3,4} = 50/1$. What is the minimum allowable input CM level if I_{SS} requires at least 0.4 V across it? Using this value for $V_{in,CM}$, calculate the maximum output voltage swing in each case.

4.15. In the circuit of Fig. 4.39(a), assume that $I_{SS} = 1$ mA and $W/L = 50/0.5$ for all the transistors.
(a) Determine the voltage gain.
(b) Calculate V_b such that $I_{D5} = I_{D6} = 0.8(I_{SS}/2)$.
(c) If I_{SS} requires a minimum voltage of 0.4 V, what is the maximum differential output swing?

4.16. Assuming that all the circuits shown in Fig. 4.44 are symmetric, sketch V_{out} as **(a)** V_{in1} and V_{in2} vary differentially from zero to V_{DD}, and **(b)** V_{in1} and V_{in2} are equal and vary from zero to V_{DD}.

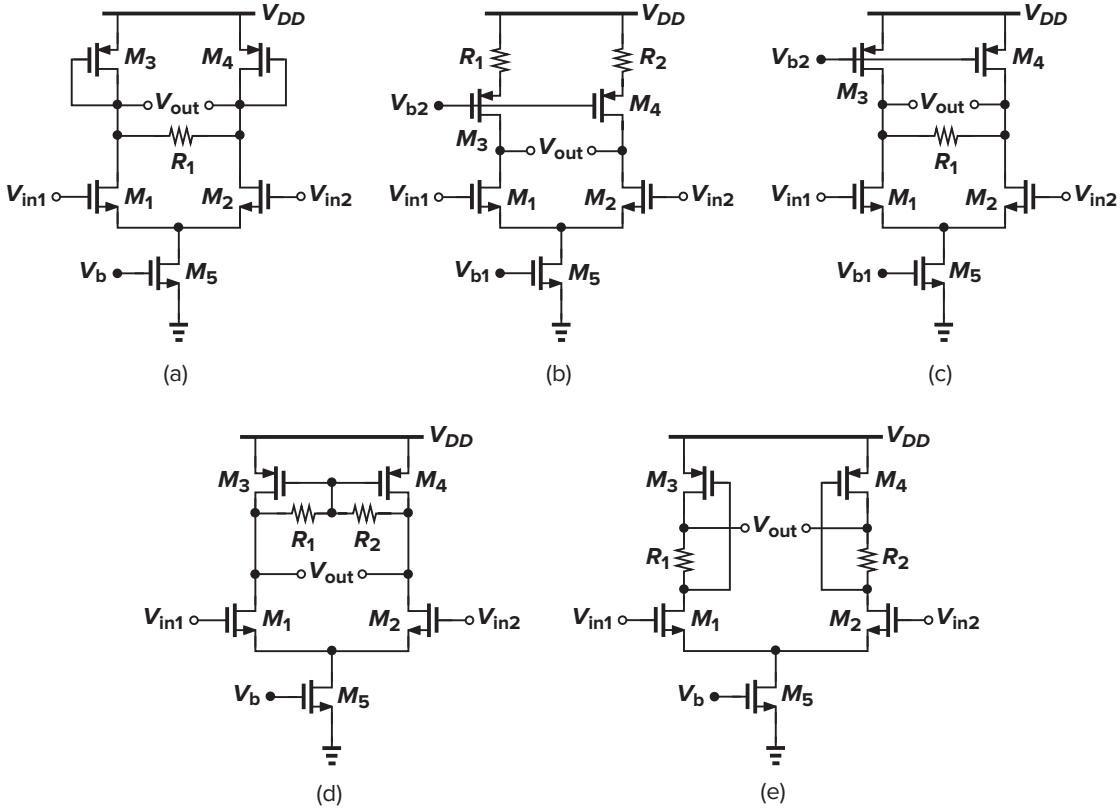


Figure 4.44

4.17. Assuming that all the circuits shown in Fig. 4.45 are symmetric, sketch V_{out} as **(a)** V_{in1} and V_{in2} vary differentially from zero to V_{DD}, and **(b)** V_{in1} and V_{in2} are equal and vary from zero to V_{DD}.

4.18. Assuming that all the transistors in the circuits of Figs. 4.44 and 4.45 are saturated and $\lambda \neq 0$, calculate the small-signal differential voltage gain of each circuit.

4.19. Consider the circuit shown in Fig. 4.46.
(a) Sketch V_{out} as V_{in1} and V_{in2} vary differentially from zero to V_{DD}.
(b) If $\lambda = 0$, obtain an expression for the voltage gain. What is the voltage gain if $W_{3,4} = 0.8W_{5,6}$?

4.20. For the circuit shown in Fig. 4.47,
(a) Sketch V_{out}, V_X, and V_Y as V_{in1} and V_{in2} vary differentially from zero to V_{DD}.
(b) Calculate the small-signal differential voltage gain.

4.21. Assuming no symmetry in the circuit of Fig. 4.48 and using no equivalent circuits, calculate the small-signal voltage gain $(V_{out})/(V_{in1} - V_{in2})$ if $\lambda = 0$ and $\gamma \neq 0$.

4.22. Due to a manufacturing defect, a large parasitic resistance has appeared between the drain and source terminals of M_1 in Fig. 4.49. Assuming $\lambda = \gamma = 0$, calculate the small-signal gain, common-mode gain, and CMRR.

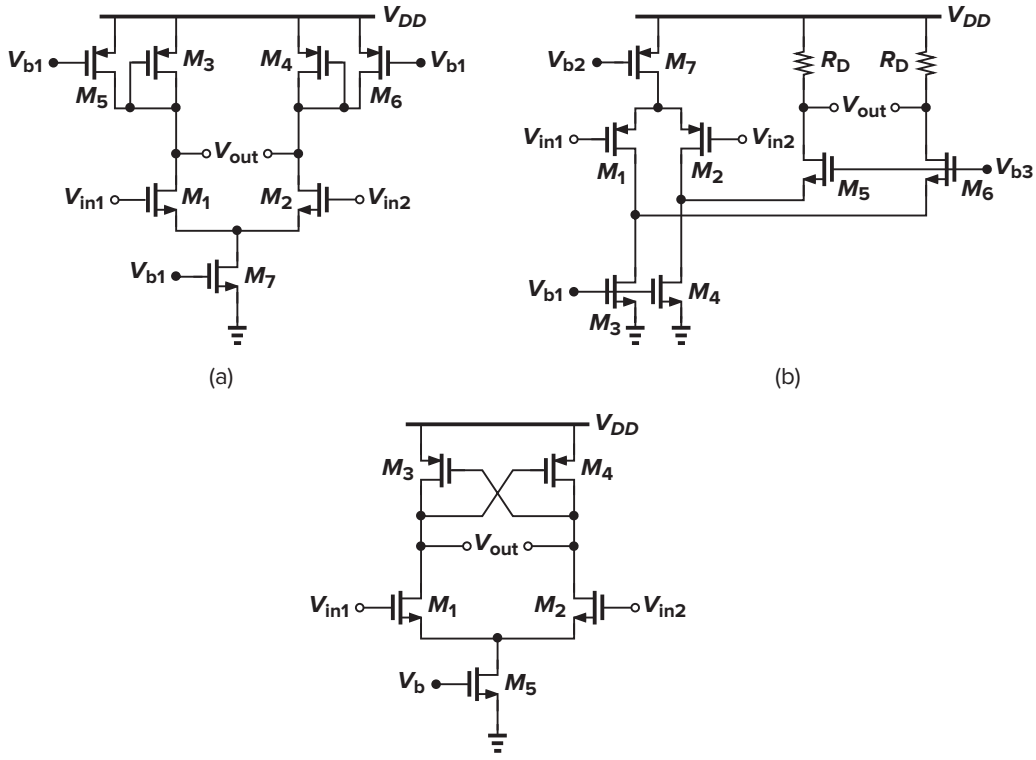


Figure 4.45

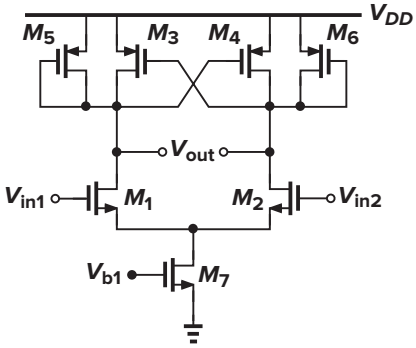


Figure 4.46

4.23. Due to a manufacturing defect, a large parasitic resistance has appeared between the drains of M_1 and M_4 in the circuit of Fig. 4.50. Assuming $\lambda = \gamma = 0$, calculate the small-signal gain, common-mode gain, and CMRR.

4.24. In the circuit of Fig. 4.51, all of the transistors have a W/L of 50/0.5, and M_3 and M_4 are to operate in the deep triode region with an on-resistance of 2 kΩ. Assuming that $I_{D5} = 20\ \mu$A and $\lambda = \gamma = 0$, calculate the input common-mode level that yields such resistance. Sketch V_{out1} and V_{out2} as V_{in1} and V_{in2} vary differentially from 0 to V_{DD}.

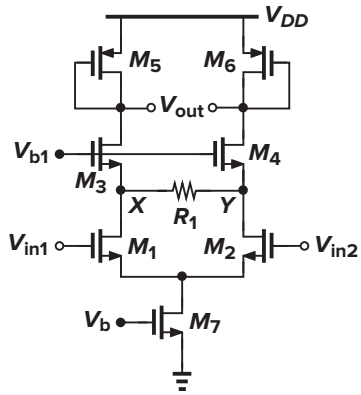


Figure 4.47

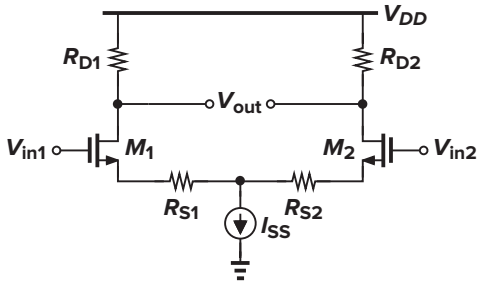


Figure 4.48

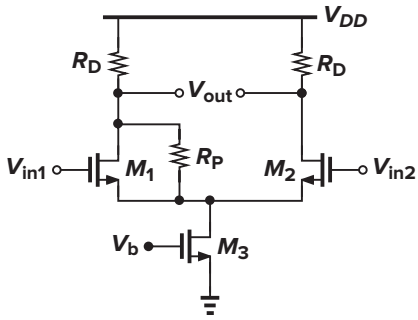


Figure 4.49

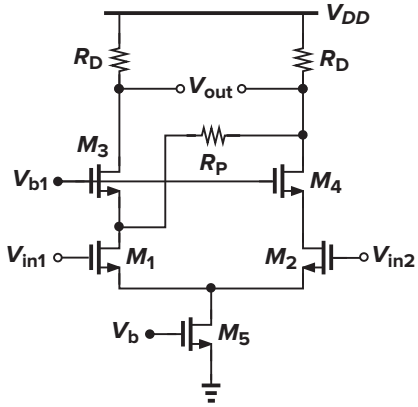


Figure 4.50

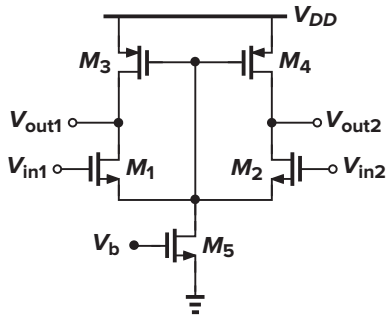


Figure 4.51

4.25. In the circuit of Fig. 4.37(b), $(W/L)_{1-4} = 50/0.5$ and $I_{SS} = 1$ mA.
(a) What is the small-signal differential gain?
(b) For $V_{in,CM} = 1.5$ V, what is the maximum allowable output voltage swing?

4.26. In the circuit of Fig. 4.39, assume that M_5 and M_6 have a small threshold voltage mismatch of ΔV and I_{SS} has an output impedance R_{SS}. Calculate the CMRR.

4.27. What happens if R_{SS} in Eq. (4.56) becomes very large? Can we obtain the same result by analyzing a differential pair having an ideal tail current source but $g_{m1} \neq g_{m2}$?

4.28. In Example 4.5, how much input dc imbalance can be tolerated if the small-signal gain must not drop by more than 5%?

4.29. In the lemma illustrated in Fig. 4.20, suppose channel-length modulation is not neglected. Assuming the two devices are connected to two equal load resistors, explain intuitively why the lemma still holds.

4.30. Does the lemma in Fig. 4.20 still hold if the devices have body effect? Explain.

4.31. Repeat Example 4.7 using Method I.

4.32. Prove the lemma illustrated in Fig. 4.20 if the tail current source is replaced by a resistor R_T.

4.33. What happens to the plots on Fig. 4.13 as W/L increases? Determine the area under the G_m plot and use the result to explain why the peak G_m must increase as W/L increases.

4.34. Assuming that I_1 and I_{SS} in Fig. 4.52 are ideal and $\lambda, \gamma > 0$, determine V_{out1}/V_{in} and V_{out2}/V_{in}.

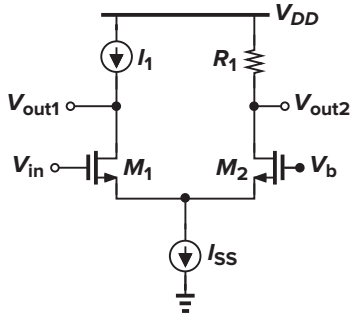


Figure 4.52

4.35. In Problem 4.11, suppose M_3 and M_4 have a threshold voltage mismatch of 1 mV. Calculate the CMRR.

Problems

Unless otherwise stated, in the following problems, use the device data shown in Table 2.1 and assume that $V_{DD} = 3$ V where necessary. All device dimensions are effective values and in microns.

5.1. In Fig. 5.2, assume that $(W/L)_1 = 50/0.5$, $\lambda = 0$, $I_{out} = 0.5$ mA, and M_1 is saturated.
(a) Determine R_2/R_1.
(b) Calculate the sensitivity of I_{out} to V_{DD}, defined as $\partial I_{out}/\partial V_{DD}$ and normalized to I_{out}.
(c) How much does I_{out} change if V_{TH} changes by 50 mV?
(d) If the temperature dependence of μ_n is expressed as $\mu_n \propto T^{-3/2}$ but V_{TH} is independent of temperature, how much does I_{out} vary if T changes from 300 °K to 370 °K?
(e) What is the worst-case change in I_{out} if V_{DD} changes by 10%, V_{TH} changes by 50 mV, and T changes from 300 °K to 370 °K?

5.2. Consider the circuit of Fig. 5.7. Assuming I_{REF} is *ideal*, sketch I_{out} versus V_{DD} as V_{DD} varies from 0 to 3 V.

5.3. In the circuit of Fig. 5.8, $(W/L)_N = 10/0.5$, $(W/L)_P = 10/0.5$, and $I_{REF} = 100\ \mu$A. The input CM level applied to the gates of M_1 and M_2 is equal to 1.3 V.
(a) Assuming $\lambda = 0$, calculate V_P and the drain voltage of the PMOS diode-connected transistors.
(b) Now take channel-length modulation into account to determine I_T and the drain current of the PMOS diode-connected transistors more accurately.

5.4. In the circuit of Fig. 5.11, sketch V_{out} versus V_{DD} as V_{DD} varies from 0 to 3 V.

5.5. Consider the circuit of Fig. 5.12(a), assuming $(W/L)_{1-3} = 40/0.5$, $I_{REF} = 0.3$ mA, and $\gamma = 0$.
(a) Determine V_b such that $V_X = V_Y$.
(b) If V_b deviates from the value calculated in part (a) by 100 mV, what is the mismatch between I_{out} and I_{REF}?
(c) If the circuit fed by the cascode current source changes V_P by 1 V, how much does V_Y change?

5.6. The circuit of Fig. 5.18(b) is designed with $(W/L)_{1,2} = 20/0.5$, $(W/L)_{3,0} = 60/0.5$, and $I_{REF} = 100\ \mu$A.
(a) Determine V_X and the acceptable range of V_b.
(b) Estimate the deviation of I_{out} from 300 μA if the drain voltage of M_3 is higher than V_X by 1 V.

5.7. The circuit of Fig. 5.23(a) is designed with $(W/L)_{1-4} = 50/0.5$ and $I_{SS} = 2I_1 = 0.5$ mA.
(a) Calculate the small-signal voltage gain.
(b) Determine the maximum output voltage swing if the input CM level is 1.3 V.

5.8. Consider the circuit of Fig. 5.29(a) with $(W/L)_{1-5} = 50/0.5$ and $I_{D5} = 0.5$ mA.
(a) Calculate the deviation of V_{out} from V_F if $|V_{TH3}|$ is 1 mV less than $|V_{TH4}|$.
(b) Determine the CMRR of the amplifier.

5.9. Sketch V_X and V_Y as a function of V_{DD} for each circuit in Fig. 5.52. Assume the transistors in each circuit are identical.

5.10. Sketch V_X and V_Y as a function of V_{DD} for each circuit in Fig. 5.53. Assume the transistors in each circuit are identical.

5.11. For each circuit in Fig. 5.54, sketch V_X and V_Y as a function of V_1 for $0 < V_1 < V_{DD}$. Assume the transistors in each circuit are identical.

5.12. For each circuit in Fig. 5.55, sketch V_X and V_Y as a function of V_1 for $0 < V_1 < V_{DD}$. Assume the transistors in each circuit are identical.

5.13. For each circuit in Fig. 5.56, sketch V_X and V_Y as a function of I_{REF}.

5.14. For the circuit of Fig. 5.57, sketch I_{out}, V_X, V_A, and V_B as a function of **(a)** I_{REF}, **(b)** V_b.

5.15. In the circuit shown in Fig. 5.58, a source follower using a wide transistor and a small bias current is inserted in series with the gate of M_3 so as to bias M_2 at the edge of saturation. Assuming M_0–M_3 are identical and $\lambda \neq 0$, estimate the mismatch between I_{out} and I_{REF} if **(a)** $\gamma = 0$, **(b)** $\gamma \neq 0$.

5.16. Sketch V_X and V_Y as a function of time for each circuit in Fig. 5.59. Assume the transistors in each circuit are identical.

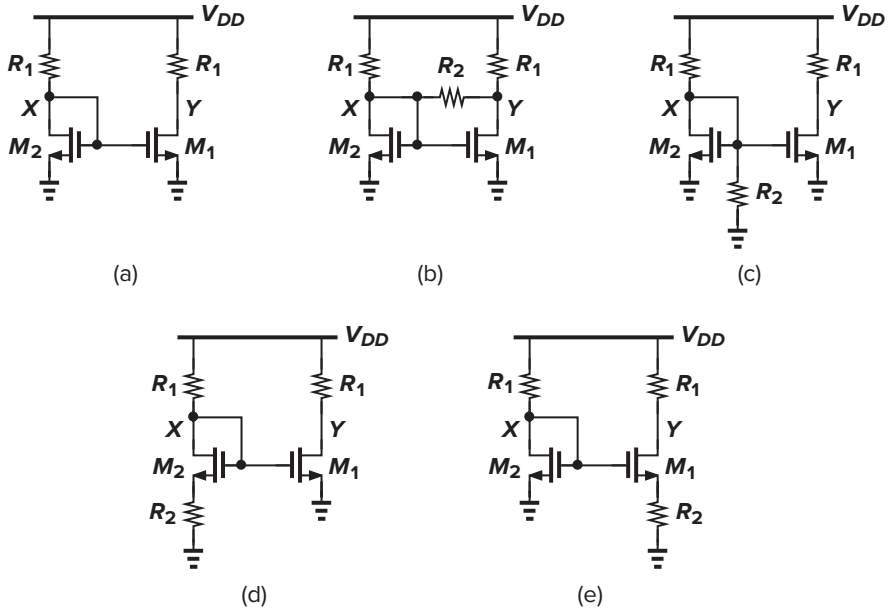


Figure 5.52

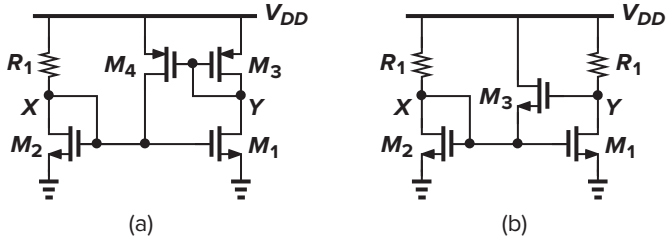


Figure 5.53

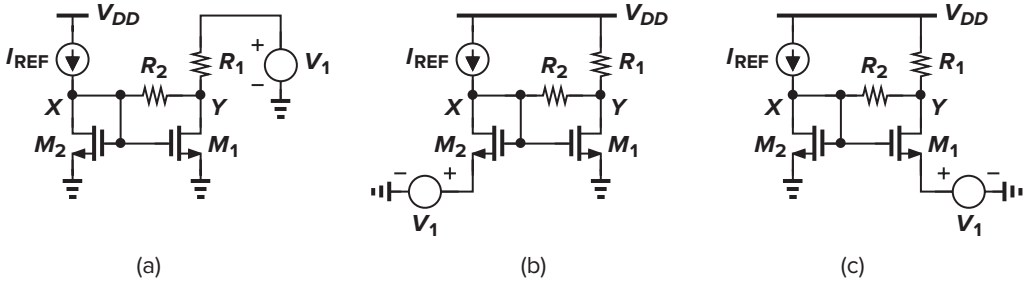


Figure 5.54

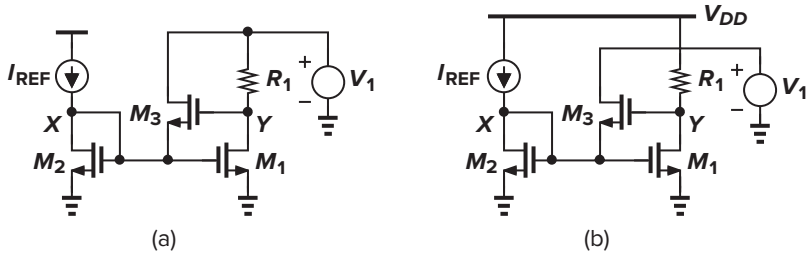


Figure 5.55

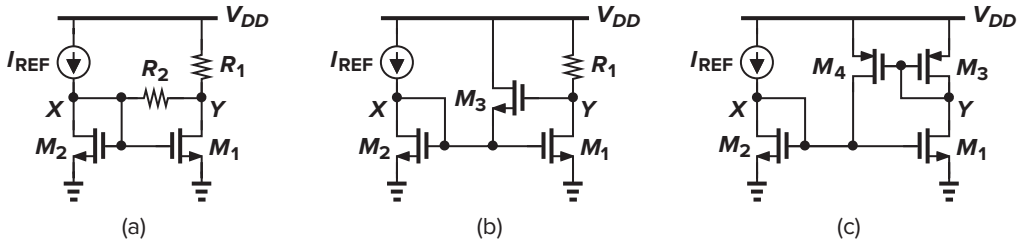


Figure 5.56

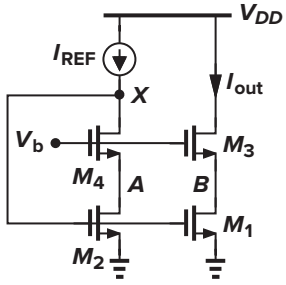


Figure 5.57

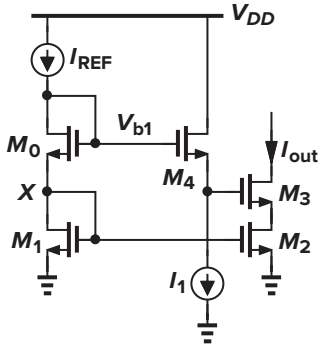


Figure 5.58

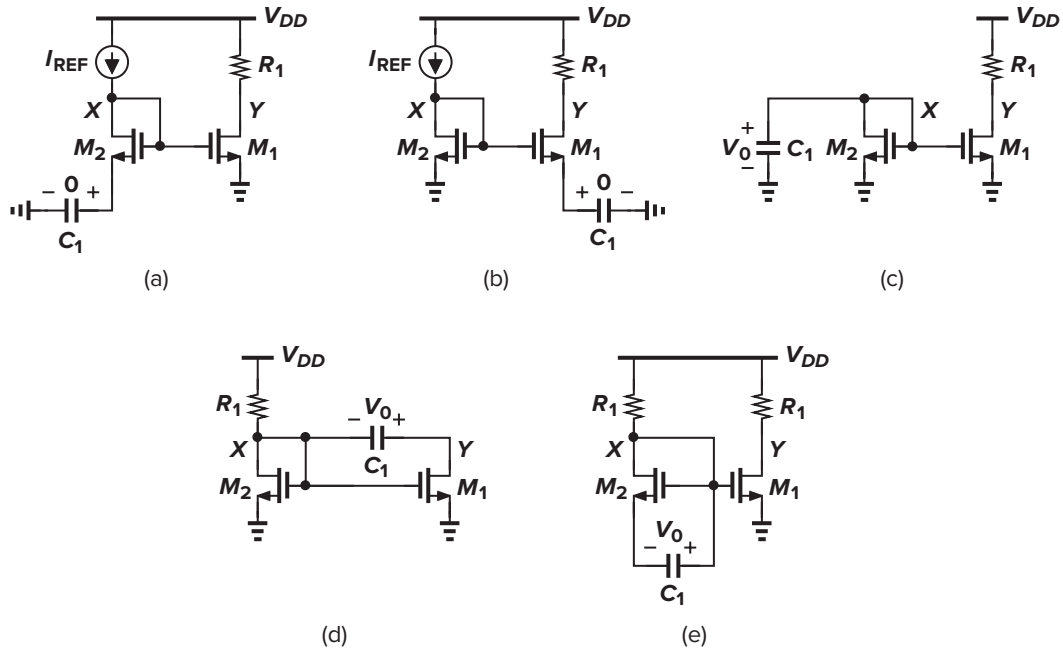


Figure 5.59

5.17. Sketch V_X and V_Y as a function of time for each circuit in Fig. 5.60. Assume the transistors in each circuit are identical.

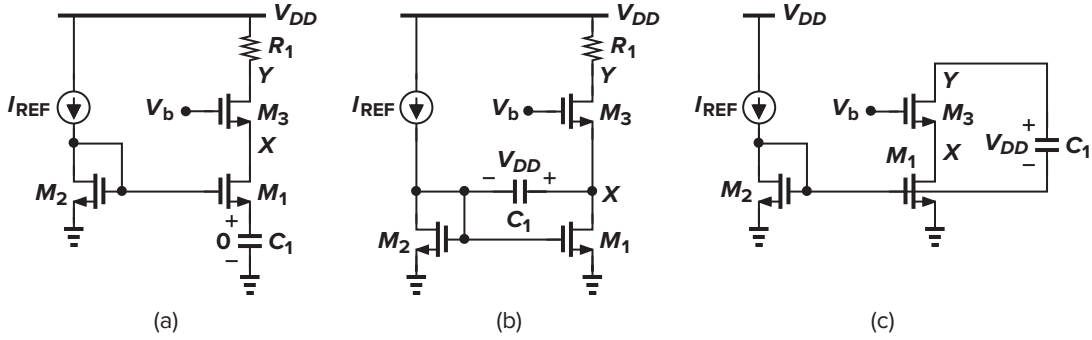


Figure 5.60

5.18. Sketch V_X and V_Y as a function of time for each circuit in Fig. 5.61. Assume the transistors in each circuit are identical.

5.19. The circuit shown in Fig. 5.62 exhibits a *negative* input inductance. Calculate the input impedance of the circuit and identify the inductive component.

5.20. Due to a manufacturing defect, a large parasitic resistance, R_1, has appeared in the circuits of Fig. 5.63. Calculate the gain of each circuit if $\lambda > 0$.

5.21. In digital circuits such as memories, a differential pair with an active current mirror is used to convert a small differential signal to a large single-ended swing (Fig. 5.64). In such applications, it is desirable that the output levels be as close to the supply rails as possible. Assuming moderate differential input swings (e.g., $\Delta V = 0.1$ V) around a common-mode level $V_{in,CM}$ and a high gain in the circuit, explain why V_{min} depends on $V_{in,CM}$.

5.22. Sketch V_X and V_Y for each circuit in Fig. 5.65 as a function of time. The initial voltage across C_1 is shown.

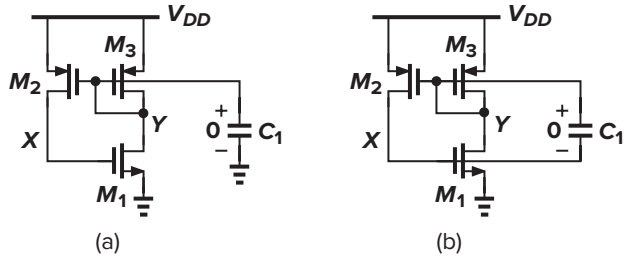


Figure 5.61

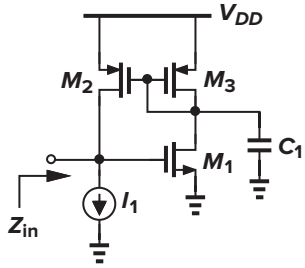


Figure 5.62

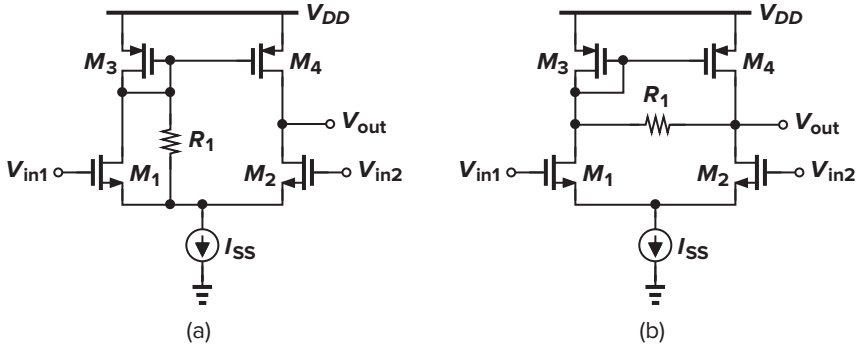


Figure 5.63

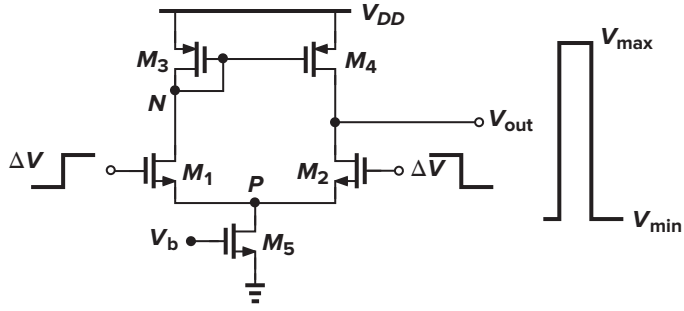


Figure 5.64

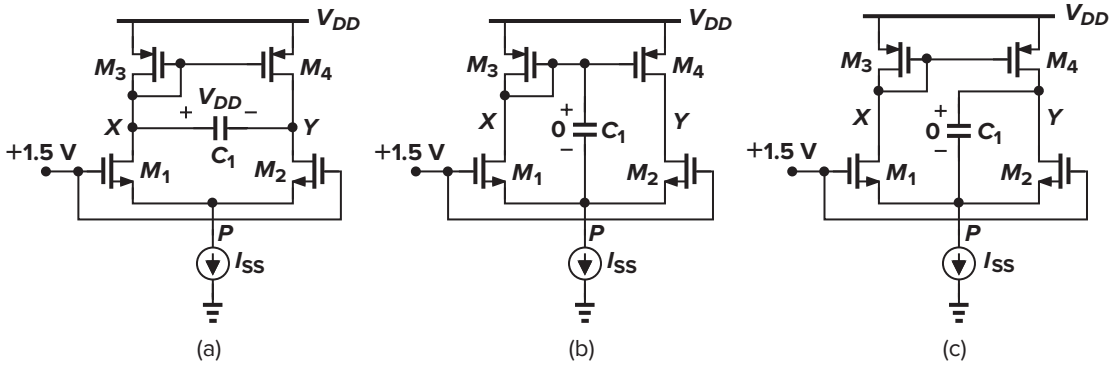


Figure 5.65

5.23. If in Fig. 5.66, ΔV is small enough that all of the transistors remain in saturation, determine the time constant and the initial and final values of V_{out}.

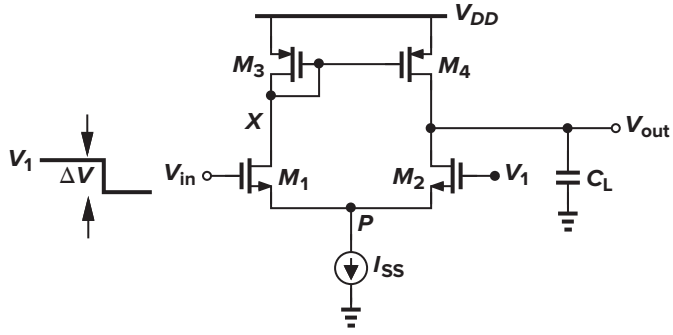


Figure 5.66

5.24. For a device operating in the subthreshold region, we have

$$I_D = \mu C_d \frac{W}{L} V_T^2 \left(\exp \frac{V_{GS} - V_{TH}}{V_T} \right) \left(1 - \exp \frac{-V_{DS}}{V_T} \right) \tag{5.64}$$

(a) If the device is in the deep triode region, $V_{DS} \ll V_T$. Using $\exp(-\epsilon) \approx 1 - \epsilon$, determine the on-resistance.
(b) If the device is in saturation, $V_{DS} \gg V_T$. Compute the transconductance.
(c) Find the relation between $g_{m,B}$ and $R_{on,R}$ in Fig. 5.43(d) using the above results.

5.25. Determine the corner frequency resulting from C_{in} in Fig. 5.47(d). For simplicity, assume C_1 is a short circuit.

5.26. Determine the supply rejection of the circuit shown in Fig. 5.67.

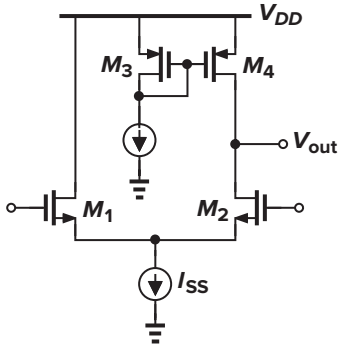


Figure 5.67

Note the duality between this expression and $Z_1 = (1 - V_Y/V_X)Z$. We also have

$$Y_2 = \frac{Y}{1 - I_1/I_2} \tag{6.132}$$

References

[1] R. D. Middlebrook, "Null Double Injection and the Extra Element Theorem," *IEEE Trans. Circuits and Systems,* vol. 32, pp. 167–180, Aug. 1989.

Problems

Unless otherwise stated, in the following problems, use the device data shown in Table 2.1 and assume that $V_{DD} = 3$ V where necessary. Also, assume that all transistors are in saturation. All device dimensions are effective values and in microns.

6.1. In the circuit of Fig. 6.3(c), suppose the amplifier has a finite output resistance R_{out}.
(a) Explain why the output jumps *up* by ΔV before it begins to go down. This indicates the existence of a zero in the transfer function.
(b) Determine the transfer function and the step response without using Miller's theorem.

6.2. Repeat Problem 6.1 if the amplifier has an output resistance R_{out} and the circuit drives a load capacitance C_L.

6.3. The CS stage of Fig. 6.13 is designed with $(W/L)_1 = 50/0.5$, $R_S = 1$ kΩ, and $R_D = 2$ kΩ. If $I_{D1} = 1$ mA, determine the poles and the zero of the circuit.

6.4. Consider the CS stage of Fig. 6.16, where I_1 is realized by a PMOS device operating in saturation. Assume that $(W/L)_1 = 50/0.5$, $I_{D1} = 1$ mA, and $R_S = 1$ kΩ.
(a) Determine the aspect ratio of the PMOS transistor such that the maximum allowable output level is 2.6 V. What is the maximum peak-to-peak swing?
(b) Determine the poles and the zero.

6.5. A source follower employing an NFET with $W/L = 50/0.5$ and a bias current of 1 mA is driven by a source impedance of 10 kΩ. Calculate the equivalent inductance seen at the output.

6.6. Neglecting other capacitances, calculate the input impedance of each circuit shown in Fig. 6.58.

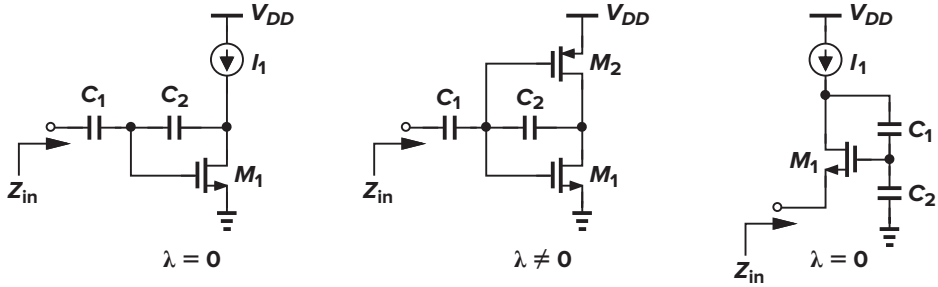


Figure 6.58

6.7. Estimate the poles of each circuit in Fig. 6.59.

6.8. Calculate the input impedance and the transfer function of each circuit in Fig. 6.60.

6.9. Calculate the gain of each circuit in Fig. 6.61 at very low and very high frequencies. Neglect all other capacitances and assume that $\lambda = 0$ for circuits (a) and (b) and $\gamma = 0$ for all of the circuits.

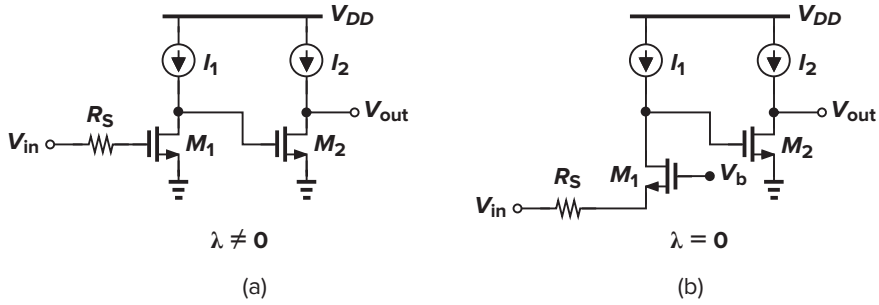


Figure 6.59

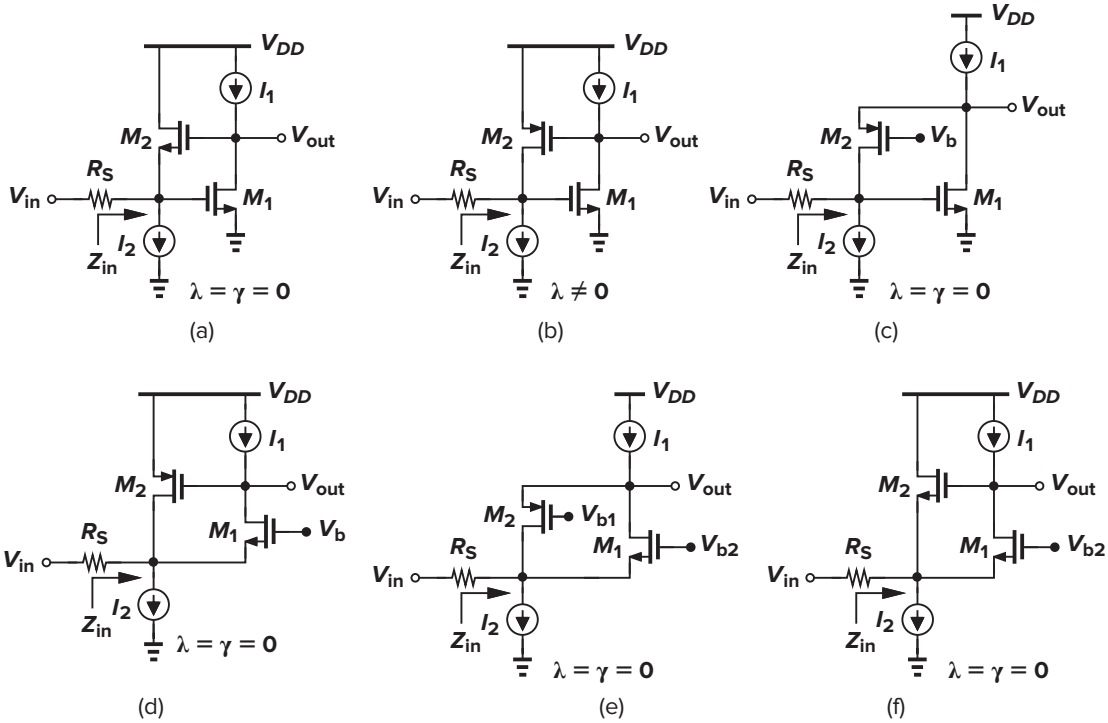


Figure 6.60

6.10. Calculate the gain of each circuit in Fig. 6.62 at very low and very high frequencies. Neglect all other capacitances and assume that $\lambda = \gamma = 0$.

6.11. Consider the cascode stage shown in Fig. 6.63. In our analysis of the frequency response of a cascode stage, we assumed that the gate-drain overlap capacitance of M_1 is multiplied by $g_{m1}/(g_{m2} + g_{mb2})$. Recall from Chapter 3, however, that with a high resistance loading the drain of M_2, the resistance seen looking into the source of M_2 can be quite high, suggesting a much higher Miller multiplication factor for C_{GD1}. Explain why C_{GD1} is still multiplied by $1 + g_{m1}/(g_{m2} + g_{mb2})$ if C_L is relatively large.

6.12. Neglecting other capacitances, calculate Z_X in the circuits of Fig. 6.64. Sketch $|Z_X|$ versus frequency.

6.13. The common-gate stage of Fig. 6.31 is designed with $(W/L)_1 = 50/0.5$, $I_{D1} = 1$ mA, $R_D = 2$ kΩ, and $R_S = 1$ kΩ. Assuming $\lambda = 0$, determine the poles and the low-frequency gain. How do these results compare with those obtained in Problem 6.9?

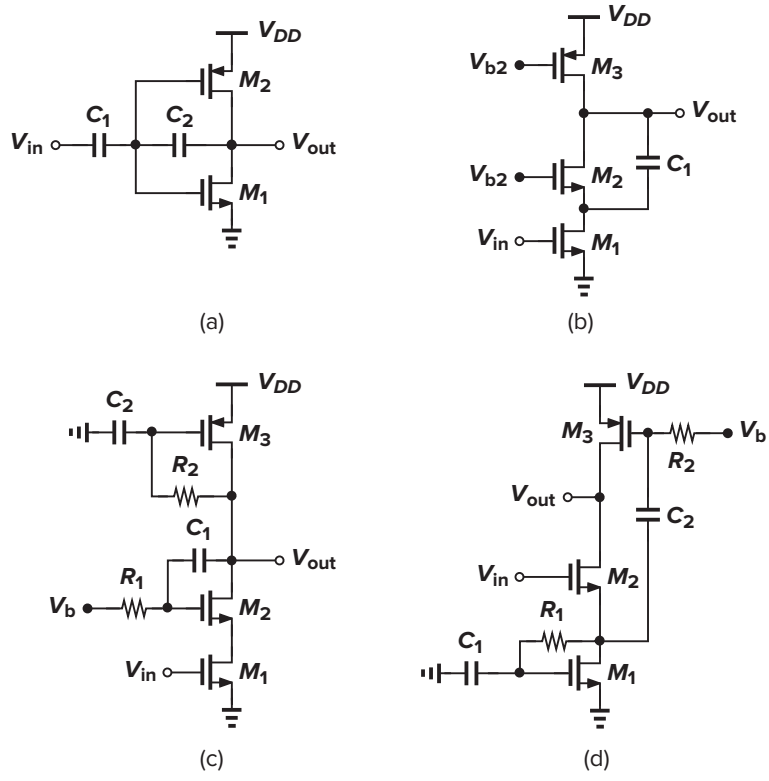


Figure 6.61

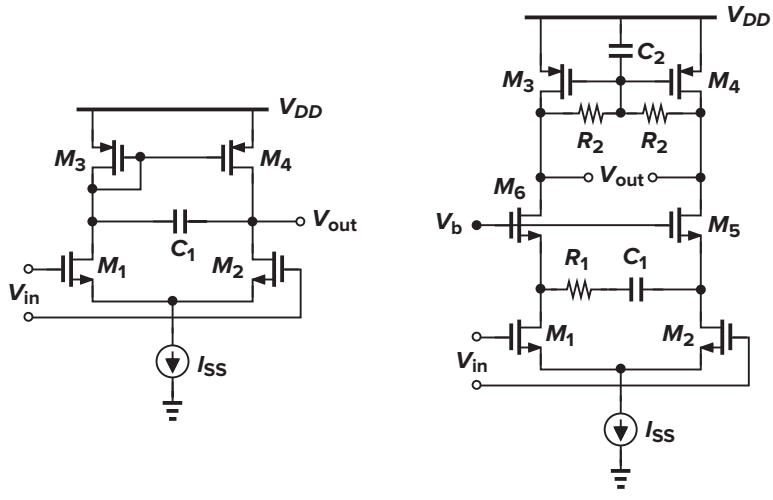


Figure 6.62

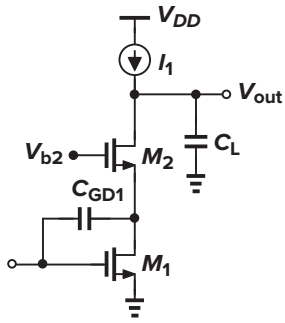


Figure 6.63

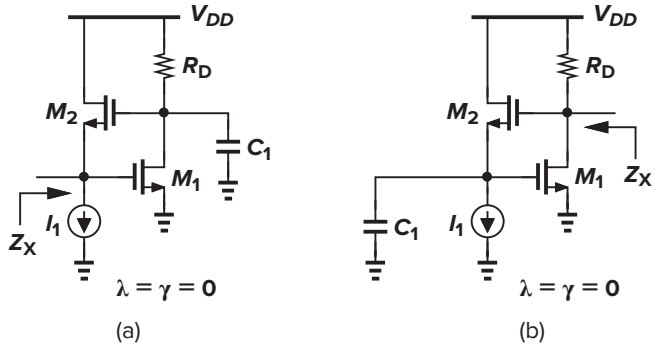


Figure 6.64

6.14. Suppose that in the cascode stage of Fig. 6.34, a resistor R_G appears in series with the gate of M_2. Including only C_{GS2}, neglecting other capacitances, and assuming $\lambda = \gamma = 0$, determine the transfer function.

6.15. Apply the method of Fig. 6.18 to the circuit of Fig. 6.41(b) to determine the zero of the transfer function.

6.16. The circuit of Fig. 6.42(a) is designed with $(W/L)_{1,2} = 50/0.5$ and $(W/L)_{3,4} = 10/0.5$. If $I_{SS} = 100\ \mu$A, $K = 2$, $C_L = 0$, and R_D is implemented by an NFET having $W/L = 50/0.5$, estimate the poles and zeros of the circuit. Assume the amplifier is driven by an ideal voltage source.

6.17. A differential pair driven by an ideal voltage source is required to have a total phase shift of 135° at the frequency where its gain drops to unity.

(a) Explain why a topology in which the load is realized by diode-connected devices or current sources does not satisfy this condition.

(b) Consider the circuit shown in Fig. 6.65. Neglecting other capacitances, determine the transfer function. Explain under what conditions the load exhibits an inductive behavior. Can this circuit provide a total phase shift of 135° at the frequency where its gain drops to unity?

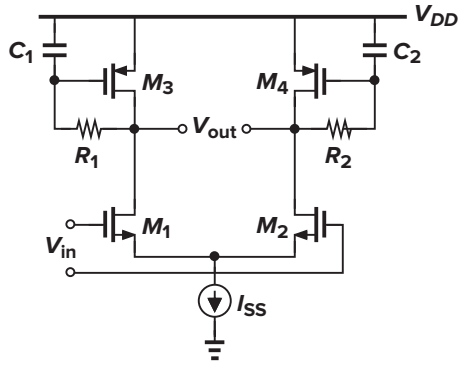


Figure 6.65

6.18. Repeat Example 6.3, but assume that I_1 is replaced with a resistor R_1.

6.19. A resistively-degenerated common-source stage bootstraps C_{GS} in a manner similar to a source follower. Estimate the input capacitance of such a stage.

6.20. Determine the transfer function of a CG stage with a resistance R_G in series with the gate, including only C_{GS} and C_{GD}. Assume $\lambda = \gamma = 0$.

6.21. Determine the transfer function of a CG stage with a resistance R_G in series with the gate, including only C_{GD} and C_{DB}. Assume $\lambda = \gamma = 0$.

6.22. Determine the transfer function of a differential pair with current-source loads for differential signals. Assume that each input is driven by a series resistance of R_S.

6.23. Consider a circuit containing only one capacitor, C_1. We set the main input to zero and apply a current source, I_X, in parallel with C_1, obtaining the voltage across it, V_X, and hence $V_X(s)/I_X(s)$ (Fig. 6.66). This impedance has the same pole as the main transfer function. Prove that the pole is given by $1/(R_1C_1)$, where R_1 is the resistance seen by C_1.

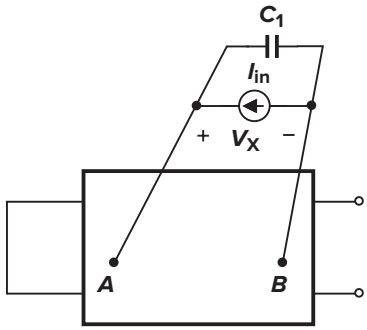


Figure 6.66

6.24. Repeat Example 6.22, but with $\lambda > 0$ and $\gamma > 0$.

6.25. Prove that the -3-dB bandwidth of N first-order identical gain stages is given by $\sqrt{\sqrt[N]{2}-1}\omega_p$, where ω_p denotes the pole of one stage.

6.26. Prove that if $C_{GD} = 0$, then Eq. (6.30) reduces to the product of two transfer functions that can simply be obtained by association of poles with the input and output nodes.

Replacing for $I_{n,in}$ from above, we obtain

$$V_{n,X} = V_{n,in} \tag{7.153}$$

That is, $I_{n,in}$ simply serves to keep $V_{n,X}$ (with respect to ground) equal to $V_{n,in}$ for different values of Z_S. This interesting result helps simplify the analysis.

Based on this observation, we modify the arrangement to that in Fig. 7.68(b), where Z_{in} simply loads the preceding stage but $I_{n,in}$ is absent. Here, too, we have $V_{n,X} = V_{n,in}$. Thus, in circuits whose output noise voltage is a weak function of the input termination, $I_{n,in}$ can be omitted if an impedance equal to Z_{in} is used to load the preceding stage.

If the above condition for $V_{n,out}$ does not hold, we may simply consider the preceding stage as part of the circuit and view the two stages as one entity. For example, the amplifier shown in Fig. 7.69 can be modeled as one stage with input-referred noise sources $V_{n,in}$ and $I_{n,in}$, thereby avoiding the complications associated with the second stage's noise voltage and current.

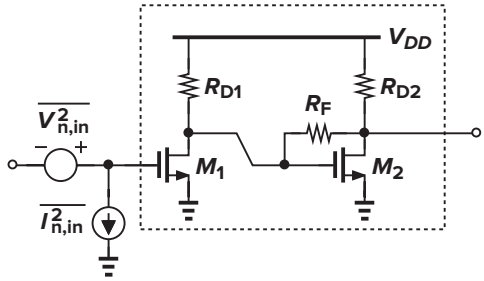


Figure 7.69 Viewing a cascade as a single circuit.

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[3] B. Razavi, Y. Ran, and K. F. Lee, "Impact of Distributed Gate Resistance on the Performance of MOS Devices," *IEEE Trans. Circuits and Systems, Part I,* pp. 750–754, Nov. 1994.

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[5] A. A. Abidi, "High-Frequency Noise Measurements on FETs with Small Dimensions," *IEEE Trans. Electron Devices,* vol. 33, pp. 1801–1805, Nov. 1986.

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[7] S. Asai et al, "High-Resistance Resistor Consisting of a Subthreshold CMOS Differential Pair," *IEICE Trans. Electronics,* vol. E93, pp. 741–746, June 2010.

Problems

Unless otherwise stated, in the following problems, use the device data shown in Table 2.1 and assume that $V_{DD} = 3$ V where necessary. Also, assume that all transistors are in saturation.

7.1. A common-source stage incorporates a 50-μm/0.5-μm NMOS device biased at $I_D = 1$ mA along with a load resistor of 2 kΩ. What is the total input-referred thermal noise voltage in a 100-MHz bandwidth?

7.2. Consider the common-source stage of Fig. 7.42. Assume that $(W/L)_1 = 50/0.5$, $I_{D1} = I_{D2} = 0.1$ mA, and $V_{DD} = 3$ V. If the contribution of M_2 to the input-referred noise voltage (not voltage squared) must be one-fifth of that of M_1, what is the maximum output voltage swing of the amplifier?

7.3. Using the distributed model of Fig. 7.21(c) and ignoring the channel thermal noise, prove that, for gate noise calculations, a distributed gate resistance of R_G can be replaced by a lumped resistance equal to $R_G/3$. (Hint: model the noise of R_{Gj} by a series voltage source and calculate the total drain noise current. Watch for correlated sources of noise.)

7.4. Prove that the output noise current of Fig. 7.39(c) is given by Eq. (7.73).

7.5. Calculate the input-referred flicker noise voltage of the circuit shown in Fig. 7.70.

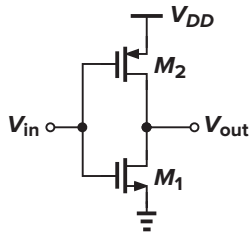


Figure 7.70

7.6. Calculate the input-referred thermal noise voltage of each circuit in Fig. 7.71. Assume that $\lambda = \gamma = 0$.

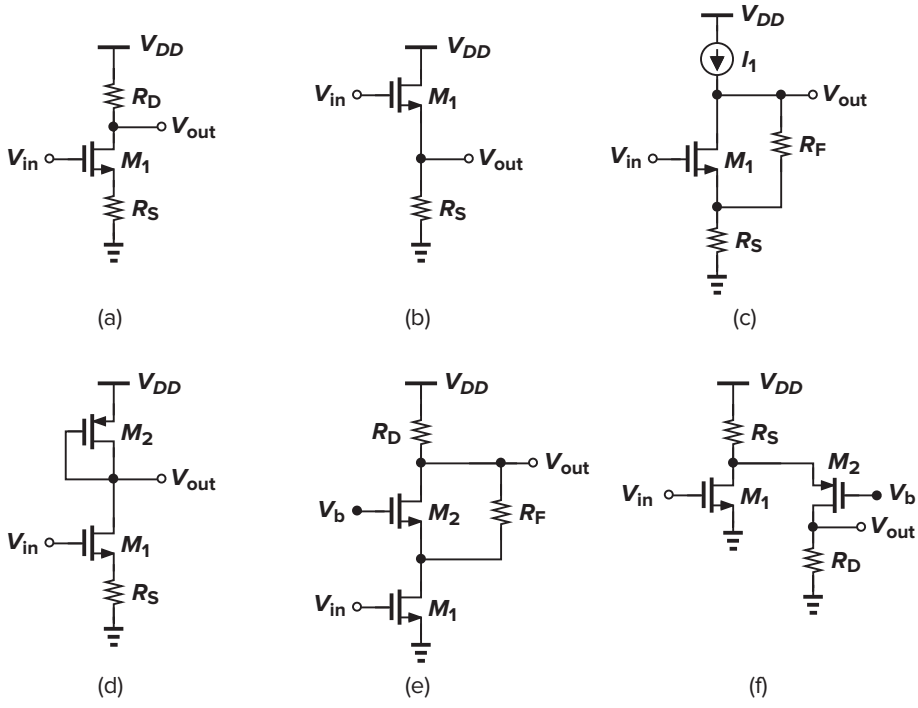


Figure 7.71

7.7. Calculate the input-referred thermal noise voltage of each circuit in Fig. 7.72. Assume that $\lambda = \gamma = 0$.